

Pearls AQI Predictor

End-to-End AQI Forecasting and MLOps System

1. Introduction

The **Pearls AQI Predictor** is an end-to-end machine-learning system developed to monitor current air quality in Karachi and forecast the **Air Quality Index (AQI) for the next 24, 48, and 72 hours**. The project follows the required workflow of data collection, feature engineering, model training, automated pipeline execution, and deployment through a web application.

The implemented system combines real-time environmental APIs, historical backfilling, Hopsworks Feature Store and Model Registry, automated GitHub Actions workflows, SHAP-based explainability, AQI health alerts, and a deployed Streamlit dashboard.

Open-Meteo APIs



Feature Engineering



Hopsworks Feature Store



Model Training & Evaluation



Hopsworks Model Registry



Prediction Pipeline



Streamlit Dashboard

2. Data Collection and Feature Engineering

Hourly air-quality and weather data for Karachi was collected using **Open-Meteo APIs**.

The main air-quality variables included PM2.5, PM10, carbon monoxide, nitrogen dioxide, sulphur dioxide, ozone, and US AQI. Weather features included temperature, relative humidity, surface pressure, and wind speed.

Historical data was backfilled to create the training dataset, as required for generating historical feature-target pairs.

After cleaning and feature engineering, the final dataset contained **4,896 rows and 27 columns** with no missing values.

The engineered features included:

- Time features: hour, day, day of week, month
- AQI lags: 1h, 3h, 6h, 24h
- Rolling AQI means: 3h, 6h, 24h
- AQI change
- Current pollutant and weather measurements

Three prediction targets were generated:

target_24h → AQI after 24 hours

target_48h → AQI after 48 hours

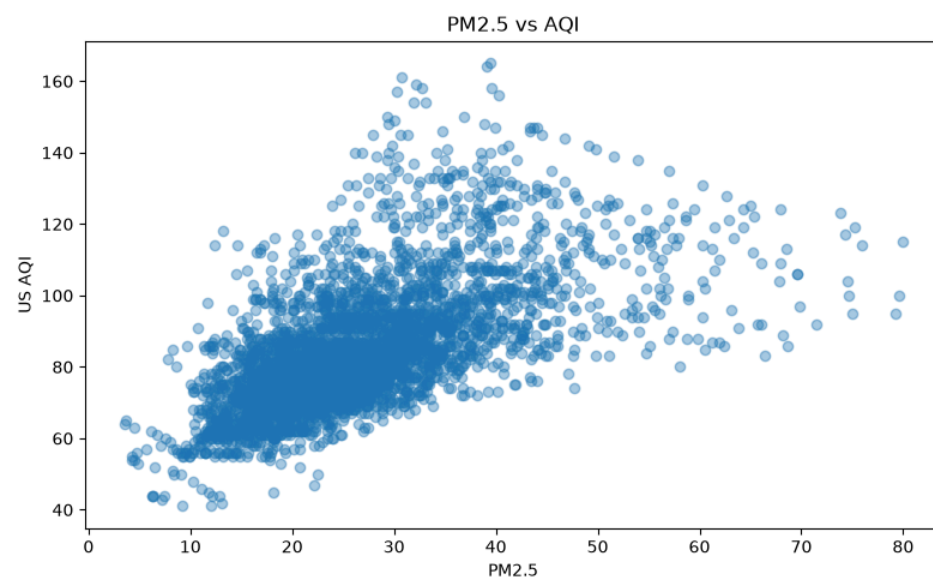
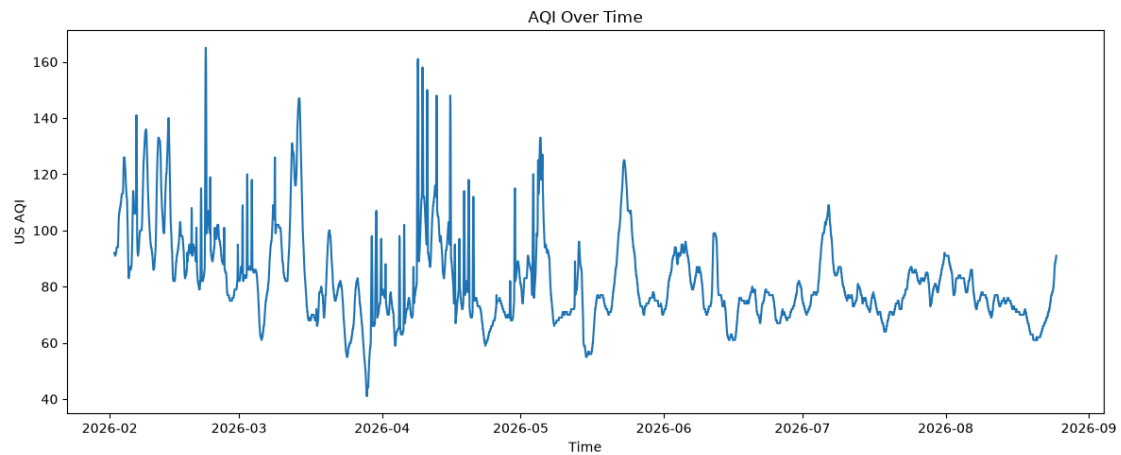
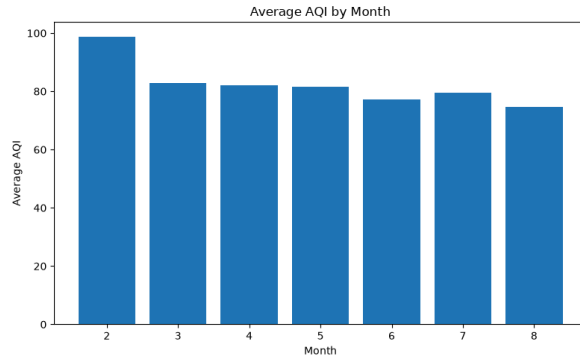
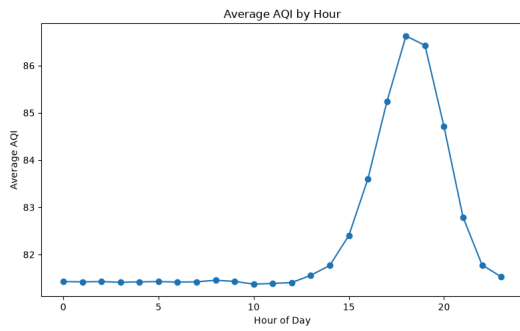
target_72h → AQI after 72 hours

This creates a **direct multi-horizon forecasting approach**, where separate models forecast each future horizon.

Exploratory Data Analysis

The dataset had a mean AQI of **82.37**, with values ranging from **41 to 165**. Recent AQI behaviour showed the strongest relationship with current AQI. The 3-hour rolling AQI mean had a correlation of **0.989**, AQI lag 1 hour had **0.987**, and PM2.5 had approximately **0.640** correlation with AQI.

EDA included AQI trends over time, average AQI by hour/month, and PM2.5 versus AQI analysis. The project brief recommends performing EDA to identify trends before forecasting.



3. Feature Store and Model Development

The engineered dataset was uploaded to the **Hopsworks Feature Store**.

Two Feature Groups were maintained:

- **karachi_aqi_features** — historical data with training targets
- **karachi_aqi_live_features** — live hourly features used during production/inference

This separates historical labelled data from live observations.

2 feature groups

show logs/traces

last created

Hudi

karachi_aqi_live_features

Latest hourly AQI, pollutant, weather and engineered features for live inference

AS

Version

2

Last updated

about 3 hours ago

Hudi

karachi_aqi_features

Hourly AQI, pollution, weather and engineered features for Karachi

AS

Version

2

Last updated

2 days ago

Three regression algorithms were evaluated:

- Ridge Regression
- Random Forest Regressor
- Gradient Boosting Regressor

A chronological 80/20 split was used to prevent future observations from leaking into training:

Training rows: 3916

Testing rows: 980

Models were evaluated using **MAE**, **RMSE**, and **R²**, as required by the project specification.

Model Results

Horizon	Best Model	MAE	RMSE	R ²
24h	Gradient Boosting	4.654	5.873	0.387
48h	Gradient Boosting	7.517	9.019	-0.291

72h Gradient Boosting 8.346 9.779 -0.481

Gradient Boosting achieved the best RMSE across all three horizons. The results also show that predictive performance decreases as the forecast horizon increases. The 24-hour model provides the strongest performance, while the negative R^2 values for 48 and 72 hours indicate that longer-horizon forecasting requires further improvement.

The best models were stored in the **Hopsworks Model Registry** as:

aqi_predictor_24h

aqi_predictor_48h

aqi_predictor_72h

MODEL COMPARISON				
=====				
forecast_horizon	model	MAE	RMSE	R2
24h	Ridge	4.774469	6.359715	0.281505
24h	RandomForest	5.109288	6.180985	0.321322
24h	GradientBoosting	4.653864	5.872581	0.387358
48h	Ridge	10.916536	14.177437	-2.189893
48h	RandomForest	8.971910	10.691317	-0.814024
48h	GradientBoosting	7.516735	9.019338	-0.291011
72h	Ridge	10.279519	13.667149	-1.893036
72h	RandomForest	8.969210	10.761738	-0.793753
72h	GradientBoosting	8.345946	9.778972	-0.481100

3 models					last created	↕
Name	Latest version	Author	Deployed versions	Description		
aqi_predictor_72h	3	AS		Best AQI forecasting model for 72h prediction horizon	View all versions	
aqi_predictor_48h	4	AS		Best AQI forecasting model for 48h prediction horizon	View all versions	
aqi_predictor_24h	4	AS		Best AQI forecasting model for 24h prediction horizon	View all versions	

4. Prediction System and Dashboard

The prediction pipeline fetches recent AQI and weather observations, creates the same features used during training, retrieves the registered models, and produces forecasts for the next 24, 48, and 72 hours.

Recent AQI + Weather



Prediction Features



Hopsworks Model Registry



24h / 48h / 72h Forecasts

A **Streamlit dashboard** was developed and deployed to present the results interactively. The application displays:

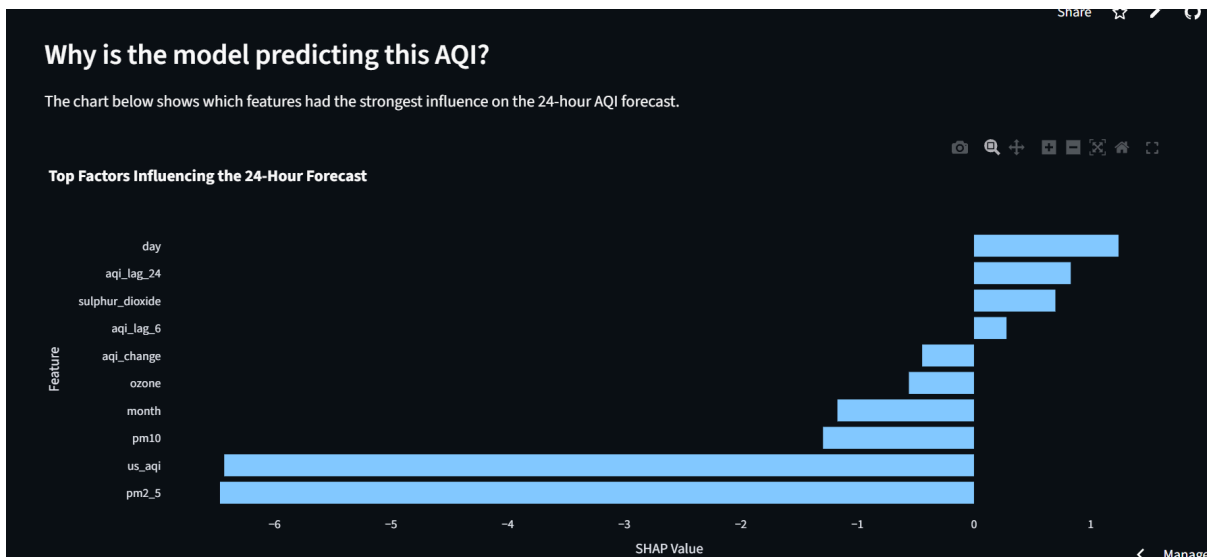
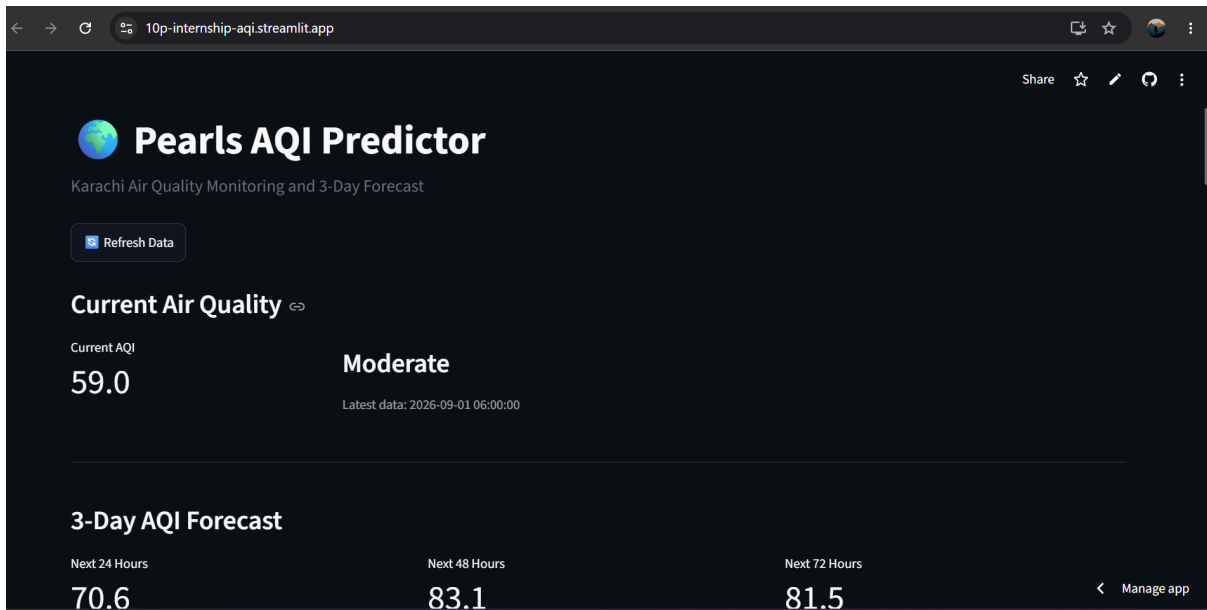
- Current AQI and AQI category
- 24h, 48h, and 72h forecasts
- AQI forecast trend
- PM2.5 and PM10 measurements
- Temperature, humidity, and wind speed
- Last updated timestamp
- AQI health alerts
- SHAP feature explanations

This satisfies the requirement for an interactive dashboard showing current and forecast AQI information.

Explainability and Alerts

SHAP is used to explain the **24-hour forecast**, showing which input features had the strongest positive or negative influence on the prediction.

The dashboard also generates AQI health notifications. For example, safe forecasts produce an informational message, while unhealthy, very unhealthy, or hazardous forecasts produce increasingly severe warnings. This implements the brief's requirements for feature explanation and hazardous AQI alerts.



i AQI is Moderate. Air quality is generally acceptable.

5. Automation, Deployment, and Conclusion

Two **GitHub Actions** workflows automate the system:

Hourly Feature Pipeline

Runs every hour:

Open-Meteo

→ Latest Feature Generation

→ Hopsworks Live Feature Group

Daily Training Pipeline

Runs daily:

Historical Feature Store

→ Model Training

→ Evaluation

→ Model Registry

This directly implements the required automation where the feature pipeline runs hourly and model training runs daily.

The Streamlit application was deployed as a web application, completing the end-to-end production workflow.

18 workflow runs		Workflow ▾	Event ▾
✓	Hourly AQI Feature Pipeline	main	
Hourly AQI Feature Pipeline #15: Scheduled			
✓	Hourly AQI Feature Pipeline	main	
Hourly AQI Feature Pipeline #14: Scheduled			
✓	Hourly AQI Feature Pipeline	main	
Hourly AQI Feature Pipeline #13: Scheduled			
✓	Daily AQI Training Pipeline	main	
Daily AQI Training Pipeline #3: Scheduled			
✓	Hourly AQI Feature Pipeline	main	
Hourly AQI Feature Pipeline #12: Scheduled			
✓	Hourly AQI Feature Pipeline	main	
Hourly AQI Feature Pipeline #11: Scheduled			

Technology Stack

Component	Technology
Programming	Python
Data Processing	Pandas, NumPy
Data Source	Open-Meteo
Machine Learning	Scikit-learn
Feature Store	Hopsworks
Model Registry	Hopsworks
Explainability	SHAP
Visualization	Plotly
Web Application	Streamlit
Automation	GitHub Actions

Limitations and Future Work

The major limitation is reduced predictive performance for the 48-hour and 72-hour horizons. Future improvements can include XGBoost or LightGBM, hyperparameter optimization, additional meteorological features, improved seasonal and cyclical features, uncertainty estimation, and multi-city forecasting.

Conclusion

The **Pearls AQI Predictor** successfully implements an end-to-end AQI forecasting and MLOps workflow. It integrates real-time and historical environmental data, engineered forecasting features, cloud-based feature and model management, automated retraining, explainable machine-learning predictions, AQI health alerts, and a deployed interactive dashboard.

The completed system therefore delivers the main required outputs: an **end-to-end AQI prediction service, automated scalable pipeline, interactive forecast dashboard, and documented model evaluation.**